

Yes, this slide is actually very relevant to your WP5.3 scope because it shows a **maturity roadmap for process anomaly detection**, starting from very simple threshold-based monitoring and gradually moving toward more intelligent process understanding.

From the screenshot, the roadmap appears to be:

Model

Capability

Model 1 Single Parameter Spatial Tolerance Band Anomaly Detection

Model 2 Single Parameter Spatial Real Tolerance Band Anomaly Detection

Model 3 Multi-Parameter Spatial Tolerance Band Anomaly Detection

Model 4 Assistance System / Non-Conformance Handling

Model 5 Situative Multi-Parameter Defect Detection

The graph is showing:

- **X-axis:** Integration phase (project maturity)
- **Y-axis:** Data skill / intelligence level
- More real parts are required as the solution becomes more sophisticated.

Model 1 — Single Parameter Spatial Tolerance Band Anomaly Detection

What they mean

Monitor one process variable only.

Example:

Tow Tension

Define:

$18 \text{ N} \leq \text{Tension} \leq 22 \text{ N}$

If tension goes outside this range:

Anomaly

Spatial means

Not only:

Time = 35 sec

but:

Location on the part

Example:

Wing Panel

↑

Anomaly happened here

They can highlight the location on the geometry.

Why start here?

Because:

- no machine model required
- no AI required
- no defect labels required

This is exactly where your Ver1 time-series framework can start.

Model 2 — Single Parameter Real Tolerance Band Detection

Difference from Model 1

Instead of a fixed threshold:

18-22 N

they learn what is normal.

Example:

At curve sections:

Expected tension = 24 N

At straight sections:

Expected tension = 20 N

Now the tolerance changes based on process context.

Why useful?

Reduces false alarms.

Instead of:

Tension > 22 = anomaly

it becomes:

Tension differs from expected local behavior

This is where your Behavioral Model starts

You are no longer doing simple monitoring.

You are learning:

Expected process behavior

from data.

Model 3 — Multi-Parameter Spatial Anomaly Detection

Now multiple variables are analyzed together.

Example:

Tension

Pressure

Speed

Temperature

Individually:

Everything looks normal

But together:

Combination is abnormal

Example:

Tension = OK

Pressure = OK

Speed = OK

But

Tension + Pressure + Speed relationship is wrong

This is impossible to detect using simple thresholds.

This is true process intelligence

This aligns strongly with:

Data-driven system identification

Behavioral modeling

Model 4 — Assistance System / Non-Conformance Handling

Now the system starts helping engineers.

Instead of:

Alarm

it provides:

Possible causes

Likely process state

Recommended investigation

Example:

Tension spike detected

Possible causes:

- Roller wear
 - Material feed issue
 - Speed mismatch
-

This matches your HMI

This is where:

Engineering insights

Expected vs Actual

Investigation dashboards

appear.

Model 5 — Situative Multi-Parameter Defect Detection

This is the highest maturity level.

Now the system knows:

Process state

Machine state

Multiple parameters

Historical behavior

Defect history

and can say:

High probability of defect

instead of:

Anomaly detected

Example:

Tow tension oscillation

+

Pressure drop

+

Specific layup region

→ likely wrinkle formation

The key message of the slide

The text at the top says:

Each new component brings you closer to an intelligent defect detection system.

Meaning:

You should NOT start with AI defect prediction.

Instead:

Step 1

Single parameter monitoring

↓

Step 2

Expected behavior monitoring



Step 3

Multi-parameter behavior understanding



Step 4

Engineering assistance



Step 5

Defect prediction

How this maps to your WP5.3 roadmap

D5 Ver1 Time-Series Analysis

Equivalent to:

Model 1

Model 2

- tension monitoring
- threshold monitoring

- anomaly detection
 - drift detection
-

D6 Ver1 Behavioral Model

Equivalent to:

Model 2

Model 3

- expected behavior
 - process twin
 - expected vs actual
 - KPI prediction
 - process-state behavior
-

D7/D11 HMI

Equivalent to:

Model 4

- anomaly dashboards
 - engineering investigation
 - process-state visualization
 - anomaly superimposition
-

Future Phase (not current scope)

Equivalent to:

Model 5

- defect prediction
 - quality prediction
 - automated diagnosis
-

For the latest MoMs you shared, I would say WP5.3 should realistically target **Model 1 → Model 4**, and **Model 5 should be considered a future evolution once real AFP defect-labelled data becomes available**.

This Nebumind roadmap is actually a very good way to think about the evolution of WP5.3.

The models are not competing approaches. They are **incremental maturity levels**.

Think of it as:

Level 1 → Detect simple deviations

Level 2 → Understand expected behavior

Level 3 → Understand process interactions

Level 4 → Assist engineers

Level 5 → Predict actual defects

Model 1 — Single Parameter Spatial Tolerance Band Anomaly Detection

What it does

Monitors one parameter against predefined limits.

Example:

Tow Tension = 20 N ± 2 N

If:

Tension > 22 N

or

Tension < 18 N

Raise anomaly.

Input

Single signal.

Example:

Tow Tension

Output

Anomaly detected

Location

Timestamp

Severity

Pros

Very simple

Can be deployed quickly.

No training data required

No defect history needed.

Easy to explain

Operators immediately understand.

Works with very little data

Even one day of data may be enough.

Cons

Too many false alarms

Example:

Tension = 23 N

might actually be acceptable at a curved section.

No context awareness

Cannot understand:

Machine state

Speed

Pressure

Temperature

No defect understanding

Only says:

Something abnormal happened

Relevance to WP5.3

Very high.

Good starting point for:

D5 Ver1 Time-Series Framework

Model 2 — Single Parameter Real Tolerance Band Detection

What it does

Learns expected behavior instead of using fixed limits.

Example:

At Location A:

Expected Tension = 20 N

At Location B:

Expected Tension = 25 N

Now anomaly becomes:

Actual $\neq$ Expected

rather than:

Actual > Threshold

Input

Single parameter.

Example:

Tow Tension

plus

Location

Process State

Output

Expected vs Actual

Deviation

Pros

Much fewer false alarms

More realistic

Captures process variability

Cons

Requires historical data

Need examples of normal production.

Still single variable

Cannot understand interactions.

Relevance to WP5.3

Very high.

This is where your:

Expected-vs-Actual Framework

starts.

Model 3 — Multi-Parameter Spatial Anomaly Detection

What it does

Looks at relationships between parameters.

Example:

Tension

Pressure

Speed

Temperature

Individually:

All OK

Together:

Combination abnormal

Example

Normal:

Speed = 500

Pressure = 20

Tension = 18

Also normal:

Speed = 700

Pressure = 28

Tension = 25

But:

Speed = 700

Pressure = 20

Tension = 25

may be abnormal.

Typical Techniques

Statistical

- PCA
- MSPC
- Mahalanobis Distance

System Identification

- ARX
- State Space

ML

- Autoencoders
 - Random Forest
-

Pros

Detects hidden process issues

Understands process relationships

Much closer to process intelligence

Cons

Requires more data

More difficult to explain

Model validation becomes important

Relevance to WP5.3

Extremely high.

This is exactly where:

Data-Driven Behavioral Model

System Identification

Process Twin

sit.

Model 4 — Assistance System / Non-Conformance Handling

What it does

Supports engineers in investigating anomalies.

Instead of:

Alarm

it says:

Possible cause

Possible impact

Relevant process variables

Example

Detected:

Tension Oscillation

System suggests:

Speed instability

Material feed issue

Roller wear

Inputs

Outputs from:

Model 1

Model 2

Model 3

Output

Engineering guidance.

Pros

Saves investigation time

Makes analytics useful

Supports operators

Cons

Requires domain knowledge

Cause-effect rules can be difficult

Relevance to WP5.3

High.

This maps directly to:

HMI

Visualization

Engineering Insights

Model 5 — Situative Multi-Parameter Defect Detection

What it does

Predicts actual defects.

Example:

Gap

Overlap

Wrinkle

Tow break

before inspection finds them.

Inputs

Process Data

Machine Data

States

History

Defect Labels

Output

Probability of defect

Example

System says:

87% probability of wrinkle formation

Typical Techniques

AI / ML

- XGBoost
 - Random Forest
 - LSTM
 - Transformers
-

Pros

Highest business value

Early defect detection

Quality prediction

Cons

Requires labeled defect data

Hard to obtain

Expensive to maintain

Difficult to explain

Relevance to WP5.3

Currently low.

Based on your MoMs:

- No clear defect labels
- No mature defect database
- Focus is process understanding

Therefore Model 5 should be considered:

Future evolution

not a core deliverable.

If I map this to your roadmap

Nebumind Model

WP5.3 Deliverable

Model 1	Ver1 Time-Series Anomaly Detection
Model 2	Expected-vs-Actual Framework
Model 3	Data-Driven Behavioral Model (System Identification)
Model 4	HMI + Engineering Analysis + Visualization
Model 5	Future Defect Prediction Framework

For the scope you have described repeatedly from the MoMs, the realistic target is:

Model 1 → Model 4

and **not Model 5 yet**. In fact, many industrial projects stop at Model 3 or Model 4 and still generate significant value because they can identify process deviations, instability, and likely causes long before they have enough data to build a reliable defect prediction AI.

Yes. After seeing the Nebumind maturity roadmap, I would actually slightly restructure your WP5.3 deliverables because it aligns very well with what we have been discussing.

The key realization is:

D5 (Time Series) $\approx$ Model 1 + Model 2

D6 (Behavior Model) $\approx$ Model 3

D7/D11 (HMI) $\approx$ Model 4

Future Scope $\approx$ Model 5

So the deliverables should progressively build toward higher process intelligence.

WP5.3 Process Intelligence & Optimization Roadmap

D1 — Requirements Gathering & Validation

Objective

Establish the functional scope, available process signals, expected outputs, anomaly definitions, process states, KPI definitions, and integration boundaries between WP5.1 and WP5.3.

Nebumind Mapping

Pre-requisite for all models.

Duration

3 Weeks

Output

- Requirements specification
 - KPI catalogue
 - Process-state catalogue
 - Signal inventory
 - Integration requirements
-

D2 — High-Level Architecture & Data Flow Design

Objective

Define the architecture supporting:

- Time-series anomaly detection
- Behavioral modeling
- HMI visualization
- Geometry mapping
- Expected-vs-actual framework

Nebumind Mapping

Foundation for Models 1–4

Duration

3 Weeks

Output

- Architecture document
 - Component architecture
 - Data flow diagrams
 - Interface definitions
-

D3 — Model-Ready Dataset Preparation (Sample mTorres Data)

Objective

Prepare two different datasets because anomaly detection and behavioral modeling require different data structures.

Dataset A

Time-series anomaly detection dataset.

Contains:

Timestamp

Tension

Pressure

Temperature

Speed

Machine State

Dataset B

Behavioral model dataset.

Contains:

Inputs

Process State

Outputs

KPIs

Quality indicators

Nebumind Mapping

Preparation for Models 1–3

Duration

4 Weeks

D4 — Data Ingestion Pipeline Development

Objective

Develop ingestion services capable of collecting, synchronizing and storing process information for analytics.

Duration

4 Weeks

Output

- Ingestion framework
- Synchronization services
- Data validation framework

D5 — Ver1 Time-Series Process Anomaly Detection Framework (Sample Data)

Nebumind Mapping

Model 1

Single Parameter Anomaly Detection

Model 2

Expected-vs-Actual Single Parameter Detection

Objective

Detect process anomalies from raw time-series signals.

Focus on:

- drift
 - spikes
 - instability
 - threshold violations
 - deviation from expected behavior
-

D5.1 Process Signal Characterization

Duration

1 Week

Study process signals and identify:

- normal operating ranges
- process variability
- process states

Output

Process signal characterization report

D5.2 Single Parameter Monitoring Framework

Duration

1.5 Weeks

Implement monitoring for:

- tension
- pressure
- temperature
- speed

using statistical thresholds.

Nebumind

Model 1

Output

Threshold-based anomaly framework

D5.3 Process Envelope Development

Duration

1.5 Weeks

Develop expected process envelopes using historical behavior.

Nebumind

Model 2

Output

Expected process behavior model

D5.4 Expected-vs-Actual Deviation Detection

Duration

2 Weeks

Implement comparison between:

Expected KPI

vs

Actual KPI

Output

Deviation detection framework

D5.5 Process Instability & Drift Detection

Duration

2 Weeks

Develop:

- drift detection
- stability monitoring
- trend monitoring

Output

Process health indicators

D6 — Ver1 Data-Driven Behavioral Model (Sample Data)

Nebumind Mapping

Model 3

Multi-Parameter Behavioral Intelligence

Objective

Learn relationships between process inputs and process outputs.

D6.1 Input–Output Mapping & KPI Definition

Duration

1 Week

Identify:

Inputs:

Tension

Pressure

Temperature

Speed

Machine State

Outputs:

Process KPIs

Quality KPIs

Process State Indicators

Output

Process I/O model

D6.2 Behavioral Model Architecture Definition

Duration

1 Week

Evaluate:

- ARX
- State Space
- NARX

- ML alternatives

Output

Model architecture selection

D6.3 Behavioral Model Training

Duration

2 Weeks

Train models using sample data.

Output

Initial behavioral model

D6.4 Multi-Parameter Behavior Learning

Duration

1.5 Weeks

Learn:

- parameter interactions
- process dependencies
- process-state effects

Nebumind

Model 3

Output

Behavioral intelligence framework

D6.5 Model Validation & Residual Analysis

Duration

1.5 Weeks

Validate:

- accuracy
- residuals
- robustness

Output

Validated behavioral model

D7 — Ver1 HMI Integration & Visualization

Nebumind Mapping

Model 4

Engineering Assistance Layer

Objective

Visualize process intelligence generated by D5 and D6.

D7.1 KPI Dashboard Development

Duration

1 Week

Display:

- KPIs
 - trends
 - process health
-

D7.2 Anomaly Dashboard Development

Duration

1.5 Weeks

Display:

- anomalies
 - severity
 - process deviations
-

D7.3 Expected-vs-Actual Visualization

Duration

1 Week

Display:

- actual KPI
 - expected KPI
 - deviation
-

D7.4 Geometry Mapping & Overlay

Duration

2 Weeks

Superimpose:

- anomalies

- KPI variations
- process events

onto AFP part geometry.

D7.5 Engineering Investigation Views

Duration

1 Week

Support:

- anomaly investigation
- process understanding

Nebumind

Model 4

D8 — Actual Data Integration & Dataset Preparation (WP5.1 Data)

Duration

4 Weeks

Rebuild:

- anomaly dataset
- behavioral model dataset

using actual AFP production data.

D9 — Ver2 Time-Series Anomaly Framework (Actual Data)

Nebumind

Advanced Model 2

Duration

6 Weeks

Enhance:

- thresholds
- process envelopes
- anomaly detection accuracy

using actual production data.

D10 — Ver2 Data-Driven Behavioral Model (Actual Data)

Nebumind

Advanced Model 3

Duration

7 Weeks

Refine:

- behavioral model
- process intelligence
- expected process behavior

using actual laying data.

D11 — Ver2 HMI & Engineering Analysis

Nebumind

Advanced Model 4

Duration

4 Weeks

Enhance:

- dashboards
- overlays
- expected-vs-actual visualization
- engineering investigation tools

using production-calibrated analytics.

D12 — Final Integration Testing & Validation

Duration

3 Weeks

Validate:

- D5 anomaly framework
- D6 behavioral model
- D7/D11 HMI
- WP5.1 integration

Output

Production-ready Process Intelligence Platform

What is NOT in current scope?

This corresponds to Nebumind Model 5:

Defect Prediction

Defect Classification

AI Quality Prediction

Automated Root Cause Diagnosis

Closed Loop Optimization

I would explicitly place this as:

Future Evolution (Post-WP5.3)

because your MoMs consistently point toward **process intelligence, anomaly detection, behavioral modeling, and visualization**, not full AI-driven defect prediction.

Good point. If your **latest scope really includes AI-based defect detection and AI-based process behavior modelling at the end of the project**, then Model 5 should not be called "future work". Instead, it becomes the **final maturity stage of WP5.3**.

The reason I previously categorized it as future work was because many of the MoMs you shared emphasized:

- anomaly detection,
- system identification,
- process understanding,
- expected-vs-actual behavior,

and there was little discussion about:

- defect labels,
- quality inspection labels,
- supervised defect datasets.

Without defect labels, Model 5 is difficult.

The key question

Do you expect to have:

Actual laying data

+

Defect records

+

Quality inspection results

by Ver2?

For example:

Part	Tensio n	Pressur e	Result
------	-------------	--------------	--------

Part 1	...	...	Good
Part 2	...	...	Wrinkle
Part 3	...	...	Gap
Part 4	...	...	Tow Break

If yes, then AI-based defect prediction becomes realistic.

If not, then AI can still be used for anomaly detection and behavioral modelling, but not necessarily defect classification.

How I would map Nebumind models now

Model 1

Rule-based anomaly detection

Thresholds

Control limits

Drift detection

Maps to:

D5 Ver1

Model 2

Expected behavior monitoring

Expected vs Actual

Process envelopes

Maps to:

D5 Ver1

D9 Ver2

Model 3

Behavioral modelling

System Identification

ARX

State Space

ML Behavioral Models

Maps to:

D6 Ver1

D10 Ver2

Model 4

Engineering intelligence

Visualization

HMI

Anomaly Investigation

Process Understanding

Maps to:

D7 Ver1

D11 Ver2

Model 5

AI-driven process intelligence

Defect prediction

Quality prediction

Advanced behavioral models

Multi-parameter learning

Maps to:

D10 Ver2 (partially)

+

New final deliverable

What I would add to your roadmap

Currently you have:

D9 Ver2 Time-Series

D10 Ver2 Behavioral Model

D11 Ver2 HMI

D12 Final Validation

I would actually split the final phase:

D9 Ver2 Time-Series Analysis

D10 Ver2 Behavioral Model

(System Identification)

D11 AI-Based Process Intelligence

(Defect Detection + Advanced Behavioral Learning)

D12 HMI Integration & Visualization

D13 Final Validation

AI can appear in two places

Track 1 — AI Behavioral Model

Instead of:

ARX

State Space

you evaluate:

Random Forest

XGBoost

LSTM

Temporal CNN

Transformer

Goal:

Predict KPI trends

Predict process states

Predict expected behavior

This is AI-based process modeling.

Track 2 — AI Defect Detection

Goal:

Predict

Gap

Overlap

Tow break

Wrinkle

Delamination indicators

from:

Tension

Pressure

Temperature

Speed

Machine states

This requires defect labels.

Looking at your current roadmap

I would now define the maturity as:

Stage 1

Statistical anomaly detection

(D5)



Stage 2

Data-driven behavioral modelling

(D6)



Stage 3

Visualization & process understanding

(D7)



Stage 4

Production-calibrated behavioural intelligence

(D10)



Stage 5

AI-driven defect detection and advanced process intelligence

(New Deliverable)



Stage 6

Final HMI integration and validation

This is actually very close to the Nebumind progression shown in the slide.

So if your project charter explicitly states:

"AI-based defect detection and AI-based process behaviour modelling shall be evaluated and demonstrated using actual laying data"

then **Model 5 is not future work**. It should become the final deliverable of WP5.3 after Ver2 datasets become available.

This is exactly where many industrial projects become confusing.

Before deciding whether you need an **AI-based defect detection deliverable**, you first need to answer:

What defects are we trying to predict?

And from everything you've shared so far, I don't think that answer is clear yet.

Let's separate three concepts

1. Process Anomaly Detection

Example:

Tow tension suddenly drops

Tow tension oscillates

Pressure spike

Temperature drift

These are:

Process anomalies

NOT defects.

A defect may or may not occur afterwards.

2. Behavioral Model (System Identification)

Behavioral model learns:

Input

↓

Process Response

Example:

Speed

Pressure

Temperature



Expected Tension

or

Speed

Pressure

Tension



Expected Process Behavior

The behavioral model itself does NOT predict defects.

It predicts:

Expected behavior

and identifies:

Deviation from expected behavior

Example

Behavioral model says:

Given:

Speed = 100

Pressure = 50

Expected Tension = 20 N

Actual:

Tension = 30 N

Then:

Deviation exists

But:

Deviation $\neq$ Defect

yet.

3. Defect Prediction

This is a completely different problem.

Now you need:

Process Data

+

Defect Labels

Example Dataset

Tension	Pressure	Speed	Result
20	50	100	Good

30	50	100	Wrinkle
18	40	100	Gap
22	60	90	Good

Now AI learns:

Process Conditions



Defect Probability

What defects could AFP predict?

This depends on what quality data exists.

Typical AFP defects include:

Tow-related

- Tow break
 - Tow drop
 - Missing tow
-

Placement defects

- Gap
 - Overlap
-

Consolidation defects

- Poor compaction
 - Voids
-

Geometric defects

- Wrinkles
 - Bridging
-

Surface defects

- Foreign object inclusion
 - Surface irregularities
-

The problem in your project

From the MoMs you've shown:

I have seen discussion about:

- tension
- pressure
- process states
- anomaly detection
- behavioral modeling

But I have NOT seen evidence of:

Defect database

or

Quality inspection labels

or

NDT results linked to process data

Therefore, what is realistic today?

You can definitely do:

D5

Process anomaly detection

Tension anomaly

Pressure anomaly

Process instability

Drift detection

D6

Behavioral modeling

Expected process behavior

Expected-vs-Actual

Process understanding

D7/D11

Visualization

Anomaly dashboard

Expected-vs-Actual dashboard

Geometry overlay

What would AI-based Process Intelligence actually mean?

Not necessarily defect prediction.

It could mean:

Option A

AI Behavioral Model

Instead of:

ARX

State Space

you use:

Random Forest

XGBoost

LSTM

Transformer

to learn:

Expected Process Behavior

This is still behavioral modeling.

Option B

Anomaly Detection AI

Instead of thresholds:

Tension > 25

use:

Isolation Forest

Autoencoder

LSTM Anomaly Detection

to find unusual process behavior.

Still not defect prediction.

Option C

True Defect Prediction

Need:

Process Data

+

Defect Labels

Then predict:

Wrinkle

Gap

Overlap

Tow Break

This is Model 5.

My recommendation

Before adding a deliverable called:

AI-Based Defect Detection

ask mTorres / plant team:

Question 1

Do we have historical defect labels?

Examples:

Gap

Overlap

Wrinkle

Tow break

linked to process data.

Question 2

Do we have inspection results?

Examples:

Manual inspection

Vision inspection

NDT inspection

Quality reports

Question 3

Can process data be linked to defect location?

Example:

This wrinkle occurred here on the part

and

These process signals generated that region

If the answer is NO

Then I would NOT create:

AI Defect Detection Deliverable

Instead I would create:

AI-Based Process Intelligence & Behavioral Modeling

where AI is used for:

- anomaly detection,
- behavioral learning,
- expected-vs-actual estimation,
- process-state understanding.

That is much more aligned with the information you've shared so far.

So before creating a D11 "AI Defect Detection" deliverable, the most important question is:

Does mTorres actually have defect-labelled AFP data available for training?

Without that, you can still do AI, but it would be AI for anomaly detection and process behavior modeling, not true defect prediction.

Yes — based on all the MoMs and scope discussions you've shared, I think this is the most important correction to make.

You should **remove any reference to defect prediction from WP5.3** unless stakeholders explicitly add it later.

From the latest scope, WP5.3 is much closer to:

Process Intelligence Platform

than:

Quality Prediction Platform

What WP5.3 is responsible for

Pillar 1 — Time-Series Process Anomaly Detection

Objective:

Identify abnormal process behavior from process signals.

Examples:

- Tension drift
- Tension oscillation
- Pressure instability
- Temperature deviations
- Process-state anomalies
- Unexpected KPI variations

Outputs:

- Anomaly events
- Process health indicators
- Instability indicators
- Process variation trends

This corresponds roughly to:

Nebumind Model 1 + Model 2

Pillar 2 — Data-Driven Behavioral Modeling (System Identification)

Objective:

Learn the expected process behavior from process data.

Examples:

Inputs

(speed, pressure, tension, process state)



Expected Process Response

Outputs:

- Expected process behavior
- Expected KPI evolution
- Expected process envelopes
- Expected-vs-actual comparison
- Process-state-aware behavior

This corresponds roughly to:

Nebumind Model 3

Pillar 3 — HMI & Process Intelligence Visualization

Objective:

Provide visualization and engineering investigation capabilities.

Examples:

- KPI dashboards
- Anomaly dashboards
- Expected-vs-actual views
- Process-state visualization
- Process activation timeline
- Geometry-based anomaly overlays
- Process variation overlays

Outputs:

- Process Intelligence HMI
- Engineering investigation tools
- Process-health visualization

This corresponds roughly to:

Nebumind Model 4

What WP5.3 is NOT responsible for

I would explicitly exclude:

Defect Prediction

Defect Classification

Quality Prediction

Closed-Loop Process Optimization

Runtime Adaptive Learning

Virtual Commissioning

Physics-Based Digital Twin

Robot Simulation

unless future MoMs add them.

Updated Deliverable Mapping

Deliverable	Scope
D1	Requirements & Validation
D2	High-Level Architecture
D3	Sample Data Preparation
D4	Data Ingestion Pipeline
D5	Ver1 Time-Series Process Anomaly Detection
D6	Ver1 Data-Driven Behavioral Model
D7	Ver1 HMI Integration & Visualization
D8	Actual Data Integration & Dataset Preparation
D9	Ver2 Time-Series Process Anomaly Detection
D10	Ver2 Data-Driven Behavioral Model
D11	HMI Architecture, Backend & Frontend Development

D12 Ver2 HMI Integration & Engineering Analysis

D13 Final Validation & Integration Testing

One-line WP5.3 Scope Statement

I would write it as:

WP5.3 develops a Process Intelligence Platform combining time-series anomaly detection, data-driven behavioral modeling (system identification), and HMI-based visualization to understand process behavior, detect process anomalies, and support engineering analysis.

That statement is fully aligned with the latest scope you've described and avoids drifting into defect prediction or optimization topics.